

Hydrocarbons

A Level Chemistry

Alkanes

Alkanes 烷烃 are saturated hydrocarbons (general formula C_nH_{2n+2}).

Making alkanes

- **hydrogenation** 氢化: add hydrogen to an **alkene** 烯烃, using a Pt or Ni catalyst and heat.
- **cracking** 裂化: break a long-chain alkane into shorter ones by heating with Al_2O_3 .

Combustion

In **combustion** 燃烧 an alkane burns in oxygen:

- **complete combustion** 完全燃烧 (plenty of oxygen) gives carbon dioxide and water.
- **incomplete combustion** 不完全燃烧 (not enough oxygen) gives water plus toxic **carbon monoxide** 一氧化碳 (CO) and soot (carbon).

Free-radical substitution

Alkanes react with chlorine or bromine by **free-radical substitution** 自由基取代, in **ultra-violet light** 紫外线. For ethane and chlorine the mechanism has three steps:

- **initiation** 引发—UV light splits the halogen into two **free radicals** 自由基: $Cl_2 \rightarrow 2 Cl\cdot$
- **propagation** 增长—radicals react and make new radicals:



- **termination** 终止—two radicals join, ending the chain: $Cl\cdot + C_2H_5\cdot \rightarrow C_2H_5Cl$

Why cracking is useful, and why alkanes are unreactive

Cracking turns heavy fractions of **crude oil** 原油 into more useful, lower- M_r alkanes and alkenes. (A **fraction** 馏分 is a group of molecules with a similar boiling-point range.)

Alkanes are generally unreactive, especially towards polar reagents. This is because the C–H and C–C bonds are strong and have little **polarity** 极性, so there is no charge to attract an attacking species.

Environmental effects

Burning alkanes in an internal combustion engine gives off carbon monoxide, oxides of nitrogen and unburnt hydrocarbons. A catalytic converter removes these by turning them into harmless gases.

Alkenes

Alkenes have a C=C double bond (general formula C_nH_{2n}). The double bond is the functional group, so alkenes are reactive.

Making alkenes

- **elimination** 消去 of HX from a **halogenoalkane** 卤代烷, using NaOH dissolved in ethanol, with heat.
- **dehydration** 脱水 of an **alcohol** 醇, using a hot Al_2O_3 catalyst or concentrated sulfuric acid.
- cracking of a longer-chain alkane.

Reactions of alkenes

Most reactions are **electrophilic addition** 亲电加成 across the double bond:

Reagent and conditions	Product
H_2 , Pt/Ni catalyst, heat	alkane
steam 水蒸气 (H_2O), H_3PO_4 catalyst	alcohol
a hydrogen halide 卤化氢 (HX), room temperature	halogenoalkane
a halogen X_2	a di-substituted alkane

There are also two oxidation reactions with acidified KMnO_4 :

- **cold, dilute** KMnO_4 adds two $-\text{OH}$ groups to give a **diol** 二醇.
- **hot, concentrated** KMnO_4 breaks the $\text{C}=\text{C}$ bond apart. The products (such as carboxylic acids or carbon dioxide) tell you where the double bond was.

Test for a $\text{C}=\text{C}$ bond

Shake the compound with orange **bromine water** 溴水. An alkene **decolourises** it (turns it colourless) by electrophilic addition. An alkane does not.

Addition polymerisation

In **addition polymerisation** 加成聚合, many alkene molecules join into one long chain, with no other product. Ethene gives poly(ethene). The long-chain product is a **polymer** 聚合物.

The mechanism and Markovnikov's rule

In electrophilic addition (for example bromine with ethene), the electron-rich $\text{C}=\text{C}$ attracts the electrophile. This forms a positive intermediate called a **carbocation** 碳正离子, which the negative part then attacks.

Alkyl groups push electrons towards the positive carbon —this is the **inductive effect** 诱导效应. So a carbocation with more alkyl groups is more stable: tertiary is more stable than secondary, which is more stable than primary. When HBr adds to propene, the more stable carbocation forms, so hydrogen adds to the carbon that already has more hydrogens. This pattern is **Markovnikov's rule** 马氏规则.